1 Correlation and Covariance

T1. Let X_i be iid with a Bernoulli distribution (with success probability = 0.6). Let $A_2 = X_1+X_2$ and $B_2 = X_1+X_3$. Find the correlation and covariance of A_2 and B_2.
Answer

$$\begin{aligned}
Cov(A_2, B_2) &= Cov(X_1 + X_2, X_1 + X_3) \\
&= Cov(X_1, X_1 + X_3) + Cov(X_2, X_1 + X_3) \\
&= Cov(X_1, X_1) + Cov(X_1, X_3) + Cov(X_2, X_1) + Cov(X_2, X_3) \\
&= Var(X_1) + 0 + 0 + 0 \\
&= 0.6(1 - 0.6) \\
&= 0.24 \\
\rho &= \frac{Cov(A_2, B_2)}{\sqrt{V(A_2)V(B_2)}} \\
&= \frac{0.24}{\sqrt{(0.48)(0.48)}} \\
&= 0.5
\end{aligned}$$

T2. Are A2 and B2 independent? Why?
Answer No, because the correlation is not zero.

T3. Given binary random variables A and B. Show that $P(A = 1, B = 1) > P(A = 1)P(B = 1)$ if $Cov(A, B) > 0$. Note that this property is only true for binary random variables or events ($P(A \cap B) > P(A)P(B)$). Hint: what is $E[AB]$ in terms of $p_a = P(A = 1) = E[A]$ and $p_b = P(B = 1) = E[B]$?
Answer

$$\begin{aligned}
Cov(A, B) &> 0 \\
E(AB) - E(A)E(B) &> 0 \\
E(AB) &> E(A)E(B) \\
P(A = 1, B = 1) &> P(A = 1)P(B = 1)
\end{aligned}$$

T4. Let $Y = -2X + E$ where X is from a discrete uniform distribution with range -1 to 1. The pmf of E is as follows:

E	-2	-1	1	2
$p(E)$	0.3	0.1	0.1	0.5

Answer
From

$$E(X) = \frac{1}{3}(-1 + 0 + 1) = 0$$

and

$$
\begin{aligned}
E(XY) &= E(-2X^2 + EX) \\
&= -2E(X^2) + E(E)E(X) \qquad \text{since X and E is independence} \\
&= -2(\frac{1}{3})(1+0+1) + 0 \\
&= -\frac{4}{3}
\end{aligned}
$$

So

$$
\begin{aligned}
Cov(X,Y) &= E(XY) - E(X)E(Y) \\
&= -\frac{4}{3}
\end{aligned}
$$

2 Fun with Bivariate Gaussians

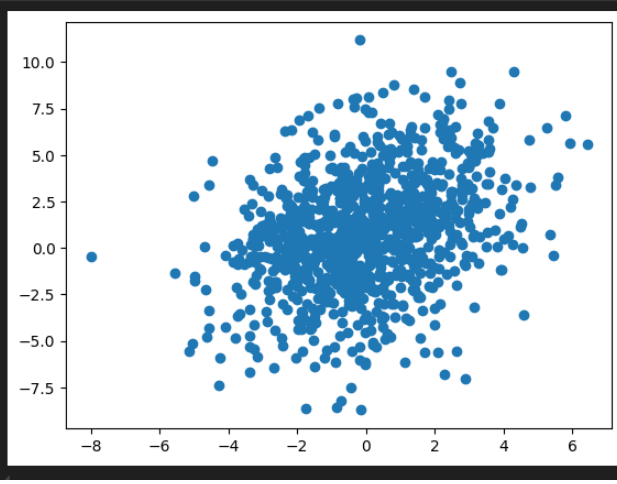
T5. Let X_1,X_2 be a bivariate Gaussian with μ = [0,1] and Σ = [[4,2],[2,9]] Compute the correlation coefficient between X_1 and X_2
Answer

$$
\begin{aligned}
\rho &= \frac{Cov(X_1, X_2)}{\sqrt{V(X_1)V(X_2)}} \\
&= \frac{2}{\sqrt{(4)(9)}} \\
&= \frac{1}{3}
\end{aligned}
$$

T6. Sample 1000 samples from this distribution using scipy.stats.multivariate_normal Plot it using a scatter plot in matplotlib. Explain how the shape of the distribution confirms your answer in the previous question.
Answer

```python
from scipy.stats import multivariate_normal
import matplotlib.pyplot as plt
xy = multivariate_normal.rvs(mean=[0,1], cov=[[4,2],[2,9]], size=1000)
plt.scatter(xy[:,0],xy[:,1])
plt.show()
```

✓ 0.1s

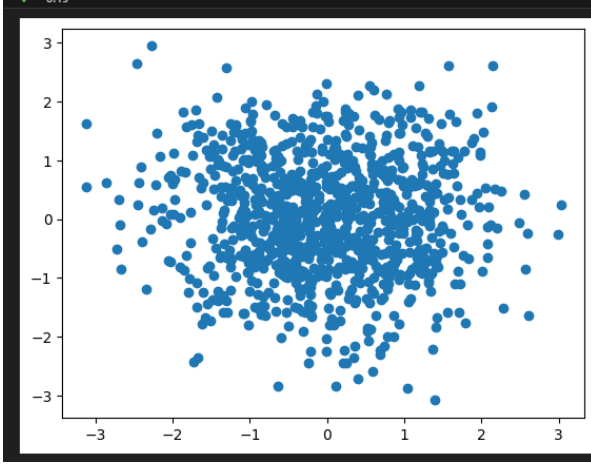


T7. Sample 1000 samples from a standard bivariate Gaussian distribution (zero mean and variance of [[1, 0],[0, 1]]). Plot the scatter plot of the data.Denote the sampled data (x_1,y_1),(x_2,y_2),(x_3,y_3),...,(x_{1000},y_{1000})

Answer

```python
from scipy.stats import multivariate_normal
import matplotlib.pyplot as plt
xy = multivariate_normal.rvs(mean=[0,0], cov=[[1,0],[0,1]], size=1000)
plt.scatter(xy[:,0],xy[:,1])
plt.show()
```

✓ 0.1s



T8. Transform the sampled data with the relation:

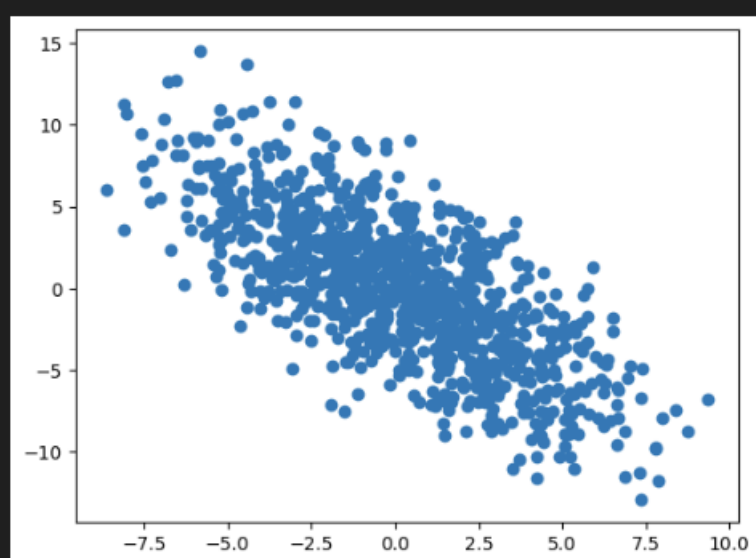
$$\begin{bmatrix} 1 & -3 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \end{bmatrix} = \begin{bmatrix} a_i \\ b_i \end{bmatrix}$$

What is the mean, covariance matrix, and correlation coefficient of A and B? Plot the scatter plot.

```python
import numpy as np
xy2 = np.matmul(np.array([[1,-3],[-4,2]]),xy.T)
mean = xy2.mean(axis=1)
cov = np.cov (xy2)
corr = cov[0,1]/(cov[0,0]*cov[1,1])**0.5
# you can also do
# corr = np.corrcoef(xy2)[0,1]
print(f'mean:{mean}')
print('covariance matrix:')
print(cov)
print('correlation coefficient:')
print(corr)
plt.scatter(xy2[0,:],xy2[1,:])
plt.show()
```

✓ 0.2s

```
mean:[0.05868879 0.01542582]
covariance matrix:
[[ 10.28026644 -10.32942371]
 [-10.32942371  20.02269838]]
correlation coefficient:
-0.7199670227915091
```



3 More fun with Bivariate Gaussians and linear algebra

T9. Prove that using this W, the covariance matrix will be an identity matrix.
Answer

The whitening transformation is defined as:

$$X' = WX$$

The covariance matrix of the transformed vector X' is:

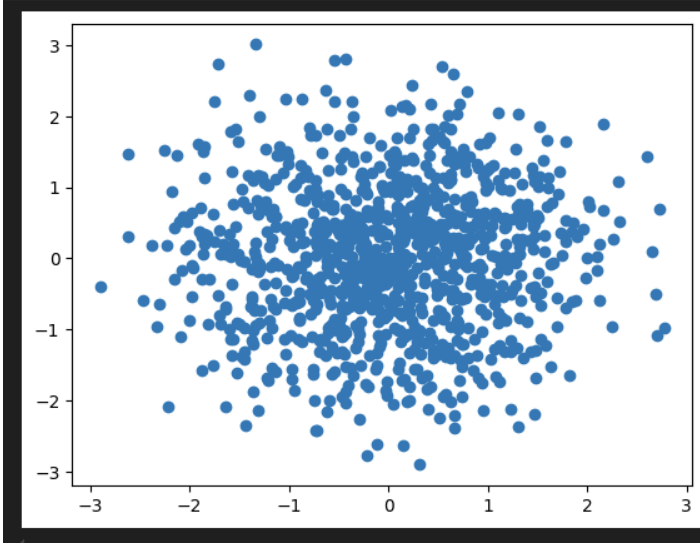
$$\begin{aligned} Cov(X') &= E(X'X'^T) \\ &= E(WXX^TW^T) \\ &= WE(XX^T)W^T \\ &= W\Sigma W^T \\ &= (\Lambda^{-\frac{1}{2}}Q^{-1})(Q\Lambda Q^T)(Q^{-1T}\Lambda^{-\frac{1}{2}T}) \end{aligned}$$

Since $Q^{-1} = Q^T$ and $Q^TQ = I$:

$$Cov(X') = \Lambda^{-\frac{1}{2}}\Lambda\Lambda^{-\frac{1}{2}} = I$$

Thus, the covariance matrix of the whitened data is the identity matrix I.

T10. Find W, apply the transform and show the scatter plot. You can use the command scipy.linalg.eigh which performs eigenvalue decomposition of a symmetric matrix.
Answer

```python
from scipy.linalg import eigh
cov = np.array([[10,-10],[-10,20]])
lamb , Q= eigh(cov)
W = np.diag(lamb**(-0.5)) @ np.linalg.inv(Q)
xy3 = np.matmul(W,xy2-xy2.mean(axis=1).reshape(-1,1))
plt.scatter(xy3[0,:],xy3[1,:])
plt.show()
```

✓ 0.1s